

A Fast Anomaly Diagnosis Approach Based on Modified CNN and Multisensor Data Fusion

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Abstract—In this article, a modified convolutional neural network (CNN) algorithm, namely 1D-GAPCNN-SVM, is proposed to address the early anomaly diagnosis problem. Considering the fact that traditional 2D-CNN based approaches contain too many model parameters and are not suitable for fast diagnosis applications using multisensor 1-D time-series measurements, 1D-CNN is introduced to deal with this problem. To reduce the number of parameters, a 1-D global average pooling layer is designed to substitute the fully connected layer with two or three layers. In order to further improve the diagnosis accuracy, a nonlinear multiclass support vector machine (SVM) is adopted to replace the traditional Softmax classifier as the final discriminator. Raw multisensor 1-D time-series data are directly fed into the diagnosis model, then the diagnosis result can be automatically generated. Two experiments, which are rolling bearing and GPS anomaly detection, have been conducted to demonstrate the effectiveness and the superior performance of the proposed method compared to the conventional SVM, K-nearest neighbor, deep neural network (DNN), and traditional 2D-CNN.

Index Terms—1-D convolutional neural network (1D-CNN), 1-D global average pooling (1D-GAP), anomaly diagnosis, GPS cyberattack, multisensor data fusion, rolling bearing, support vector machine (SVM).

I. INTRODUCTION

ANOMALY diagnosis is a key technology for reliable and safe operations of modern industrial equipment [1], [2]. As the complexity of equipment increases, anomalies are more likely to occur during the operation, which can potentially impose significant economic losses and fatal consequences [3]. In practice, early detection, isolation, and estimation of anomalies are beneficial for safe and secure operations of the complex equipment [2]. Therefore, these topics have been attracted attention in the past few decades [1]–[5]. In general, mainstream fault diagnosis approaches can be classified into three categories, namely model-based [4], [6], knowledge-based [7], [8], and data-driven based approaches [9]–[12].

Model-based approaches formulate the fault evolution process mathematically to facilitate anomaly diagnosis [2], [4]. Even though in this approach the real-time and satisfactory performance can be achieved, accurate mathematical models are difficult to obtain [5]. Besides, this kind of approach is made to address specific fault diagnosis problems. Thus, it is not straightforward to apply them to other systems. Knowledge-based approaches make decisions according to the knowledge and experience of the human experts, which makes the performance largely dependent on the accuracy and integrity of the knowledge base [7], [8]. Data-driven-based methods leverage various signal processing (SP) and statistical analysis (SA) techniques to extract key features of the anomalies for diagnosis [9], [10]. Nowadays, with the advent of emerging technologies such as “Internet +,” industrial Internet of Things, 5G communications, and smart sensing, data-driven based diagnosis methods are more commonly studied and applied [1], [2]. Data-driven methods can be classified as SA methods [11]–[13], SP methods [14]–[17], and artificial intelligence (AI) methods [18]–[21]. The SA-based methods, such as principal component analysis [11], Fisher discriminant analysis [12], and exponential discriminant analysis (EDA) [13] have been applied to fault diagnosis and anomaly detection. These methods have achieved better results in the dimensional reduction of monitoring data [12], [13]. In particular, EDA is proposed for solving the small sample size problem [13]. However, the discriminant analysis method is

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essentially still a linear classifier [13], which has a limited feature learning ability. In addition, the selection of key variables has a large effect on the discrimination accuracy, which depends on the prior knowledge of engineers. Therefore, the feature learning ability for big data samples of these methods is poor.

To extract representative features out of the raw big data for the purpose of fault diagnosis, some advanced SP-based techniques are commonly applied, such as empirical mode decomposition [14], Hilbert-Huang transform [15], fast Fourier transform (FFT), sparse filtering [16], and wavelet transform [17], etc. However, data processing of these methods extensively rely on the knowledge and experience of the operator [10]. Moreover, it is subjective and time-consuming such that many micro features from incipient faults may be ignored [18]. In recent years, with the development of AI, AI-based methods are applied in fault diagnosis domain. Examples can be found in the case of support vector machine (SVM) [14], K-nearest neighbor (KNN) [19], and artificial neural network [20]. These methods have been applied in several problems, however, these algorithms demonstrate insufficient feature extraction ability due to their shallow network structure [18]. Therefore, these methods are unable to extract deep discriminative features from raw data [3], [10], [18]. Thus, some hidden micro features are unlikely to be extracted by them [5].

To improve the diagnosis performance by extracting more detailed features, convolutional neural network (CNN) is used for anomaly diagnosis [21]–[23]. Piedad *et al.* [21] proposed a CNN-based fault diagnosis method for asynchronous motors. However, this method still uses an FFT technique to process the raw fault data, which relies on the experience and prior knowledge of engineers. Chen *et al.* [22] proposed a diagnosis method based on a cyclic spectral coherence and CNN for rolling bearing. Similarly, this method still needs prior feature extraction from raw monitoring data. Dibaj *et al.* [23] proposed a fault diagnosis approach based variational mode decomposition and CNN for rotating machinery. Although the abovementioned methods use CNN algorithms, conventional SP techniques are still employed for manual feature extraction from raw fault data in their algorithms. Long *et al.* [3] and Hoang *et al.* [24], respectively, proposed a 2D-CNN model for fault diagnosis using directly the raw monitoring signal. In their methods, to meet the input format requirements of the 2D-CNN model, the raw 1-D time-series signal is transformed into a 2-D gray image through a data reconstruction method [24]. Then, these reconstructed 2-D gray images are fed into the 2D-CNN model for fault diagnosis. However, there is a risk that the correlation information between adjacent raw data is destroyed in the data reshaping process. To maintain the correlation information, a 1D-CNN model is proposed in [25] and [26] to deal with the raw 1-D time-series data for the fault diagnosis of a motor, and a rolling bearing, respectively. Chen *et al.* [18] proposed a multiscale 1D-CNN and a long short-term memory model for bearing fault diagnosis. Wang *et al.* [27] proposed a 1D-CNN model to identify the bearing fault from the vibro-acoustic data. However, these 1D-CNN models still contain a fully connected

(FC) multilayer perceptron layer structure with 2–3 layers as a discriminative learning classifier [18], [25]–[27]. The intrinsic shortcoming of the FC layer is the excessive number of parameters to be trained [5]. The number of parameters in the FC layer is 80–90% of the total number of parameters in the entire CNN model [28], and also it makes the computation quite time-consuming, making it unsuitable for fast anomaly diagnosis.

Motivated to overcome the aforementioned limitations, in this article, a novel deep learning model, called modified 1D-GAPCNN-SVM, is developed for fast anomaly diagnosis with multisensor data. The proposed method mainly addresses three intractable problems of the current 2D-CNN model, including inefficient multisensor data fusion, slower diagnosis speed, and lower diagnosis accuracy. The procedure for implementing the proposed method, which explains our main research contributions, are listed as follows.

First, a simple and highly efficient multichannel data fusion method is introduced to solve the multisensor data fusion problem at data level. In the proposed method, raw multisensor 1-D time-series data is rapidly integrated into a 1-channel 2-D feature map.

Second, a novel fast anomaly diagnosis framework based on the modified 1D-GAPCNN-SVM is proposed, which improves the conventional 1D-CNN structure by designing a 1-D global average pooling (1D-GAP) layer to substitute a FC layer with 2–3 layers. Through this, we reduced the number of model parameters by 80%, and the training speed and detection speed of the proposed method have been improved.

Third, a nonlinear multiclass SVM is introduced to replace the ordinary Softmax function as the final classification discriminator in the model testing stage. And the fault diagnosis accuracy is further improved. The whole diagnosis process of the proposed diagnosis framework does not require any manual feature extraction operations, and it is also independent of expert experience and prior knowledge.

The rest of this article is structured as follows. In Section II, our proposed diagnosis method based on the modified 1D-GAPCNN-SVM is described in detail. In Section III, experimental results are presented and discussed. Finally, Section IV concludes this article.

II. PROPOSED METHOD FOR ANOMALY DIAGNOSIS

This article develops a novel deep learning algorithm, modified 1D-GAPCNN-SVM, for fast and early anomaly diagnosis. The model structure of the proposed algorithm is shown in Fig. 1, from which it can be observed that the proposed algorithm contains an input data fusion layer (IFL), a feature extraction layer (FEL), a 1D-GAP layer, and a logic discrimination output (LDO) layer. The proposed procedure has four steps. First, raw 1-D time-series data from multiple sensors is rapidly fused in the input layer, which performs multichannel process operations. Second, the FEL extracts the deep representative features from the fused 2-D input feature map. Third, the 1D-GAP layer performs dimensional down-sampling and parameter reduction

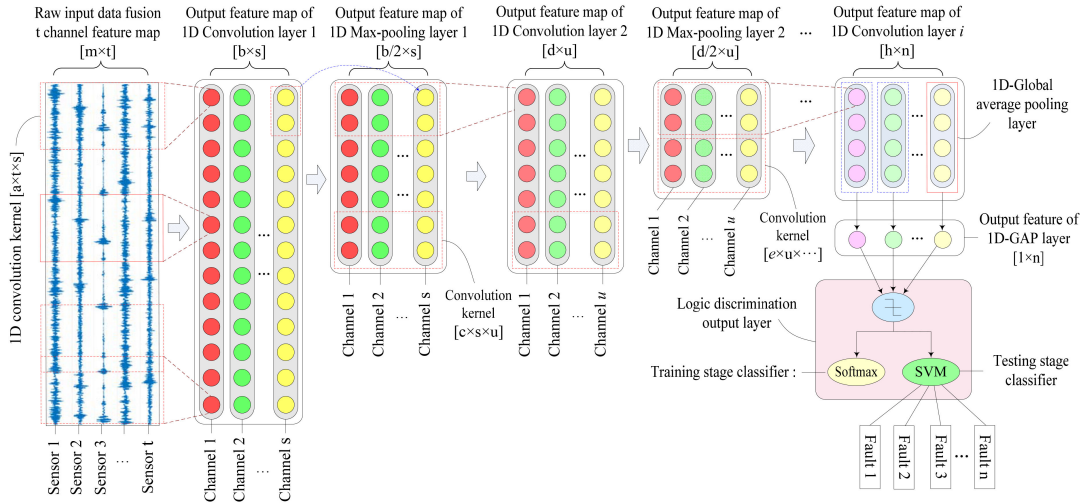


Fig. 1. Model structure of the improved 1D-GAPCNN-SVM intelligent fault diagnosis algorithm.

on the high-dimensional output data of the FEL. Finally, the result of anomaly diagnosis is produced, and misclassification information is automatically generated by the SVM classification discriminator. The end-to-end model structure has strong operability and universality.

Compared with the conventional CNN model, the main advantages of the proposed method can be summarized as follows. First, the proposed method is more efficient for multisensor 1-D time-series data fusion problems. Raw 1-D time-series data from multiple sensors can be automatically integrated into a 1-channel 2-D feature image in the IFL. Second, the proposed method is more suitable for early anomaly diagnosis and fast fault detection. The proposed method improves the traditional 1D-CNN model structure by introducing a 1D-GAP layer to replace the FC layer with 2–3 layers, which can effectively reduce the model parameter quantity and diagnosis waiting time of the traditional 1D-CNN model. Third, the proposed method can further enhance the fault diagnosis accuracy by incorporating a multiclass SVM as the final classification discriminator. The SVM can further extract the key representative feature data from the sparse output feature data of the FEL. Fourth, the improved 1D-GAPCNN-SVM method could further reduce the risk of model overfitting and avoid the gradient disappearance problem by reducing the number of network layers and trainable parameter quantity of the original 1D-CNN model.

The proposed method mainly focuses on the anomaly diagnosis problem using multiple sensors. Potential applications of the proposed method are those with continuous 1-D time-series data, such as vibration, sound, current, voltage, position, and similar monitoring signals. The proposed 1D-GAPCNN-SVM algorithm is detailed as follows.

A. Input Data Fusion Layer

The input layer has two functions, one of which is to receive and fuse the raw multichannel 1-D time-series fault data from multiple sensors, while the other one is to perform

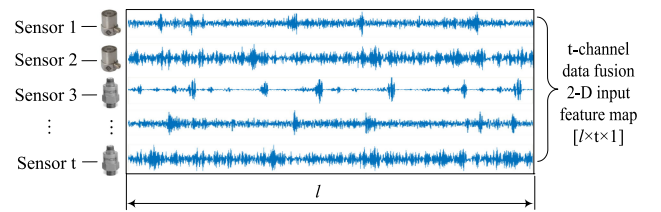


Fig. 2. Multichannel data fusion process of multiple sensors data.

some necessary data pre-processing operations [5], including data normalization, data segmentation, and data augmentation.

1) Multichannel Data Fusion: In the real-monitoring systems, multiple sensors are generally deployed for fault data collection [24]. The multichannel 1-D time-series data from multiple sensors need to be fused first. In general, sensor data fusion can be classified into three types [30]: data level, feature level, and decision level. Data level sensor fusion can achieve superior performance because loss of information is less in this method compared to the other two methods [1]. In this article, a simple and highly efficient data fusion method is proposed to solve the multichannel data fusion problem at the data level.

As shown in Fig. 2, it is assumed that each sensor can provide a raw 1-channel 1-D time-series data segment that contains l data points. In the input layer, these t -channel 1-D time-series data segments are automatically fused into a 1-channel 2-D input feature map. The size of each 1-channel 2-D feature map is $[l \times t \times 1]$, where l represents the length of the data segment, t is the number of sensors, and 1 represents the layer quantity of the 2-D feature map after data fusion.

2) Data Normalization: After the data fusion operation, different samples still have different magnitude ranges. Large magnitude variance among different fault samples is not suitable for CNN model training [29]. To solve this problem, data normalization is used in this article. In each channel, the raw data are first normalized to the range of $[0, 1]$. The common data normalization uses the max–min normalization method [5],

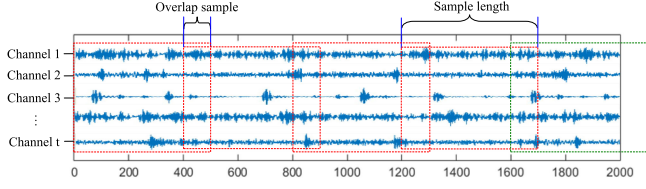


Fig. 3. Data segmentation schematic of the overlap sampling method.

given as follows:

$$X = \{x_i\} = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \quad (1)$$

where $x_{\max}$ represents the maximum value in the input sample, $x_{\min}$ represents the minimum value, and X is the result after standardization with its value between 0 and 1.

3) Data Segmentation With Overlap Sampling: After data normalization, the fault data of each channel is still a long 1-D time-series data. To obtain more training samples and to improve the correlation between the adjacent samples, a data segmentation method with the overlap sampling sliding window operation is designed, as shown in Fig. 3. In our method, each long multichannel 1-D time-series data segment can be divided into many short data segments with the same length. Each short 1-D time-series data segment is a new training fault sample.

The basic process of data segmentation with overlap sampling can be summarized as follows. First, the expected training sample length is set according to the sampling period, the fault frequency, and the execution speed. Second, the overlap sampling rate (the number of overlapping data points) can be set according to the number of expected training samples and raw monitoring data, etc. According to the previous settings, the computer program can generate a sliding sampling window, which can slide on the raw multichannel 1-D long time-series data. The multiple short time-series data segments can be saved automatically when the sliding window slides a stride. In this article, an empirical formula of sample quantity calculation for overlap sampling is proposed as follows:

$$F_{\text{num}} = \text{CEILING} \left(\frac{I_{\text{num}} - W_{\text{num}} + 1}{S_{\text{num}}}, 1 \right) \quad (2)$$

where I_{num} is the length of the raw data segment, W_{num} is the length of the sliding window, S_{num} is the number of the data points in each stride, and F_{num} represents the number of the new generated fault samples. CEILING function returns the next rounded up integer. For instance, in Fig. 3, it shows a segment of t -channel 1-D time-series data. Here, $I_{\text{num}}=2000$, $W_{\text{num}}=500$, $S_{\text{num}}=400$. So, $F_{\text{num}}=4$. Therefore, it can generate four new training fault samples. The size of each fault sample is $[500 \times t]$, and the overlap sampling rate is 20%. In addition, it is worth noting that the proposed sliding window sampling method is only used for the expansion of training data. As long as the data acquisition system obtains the data quantity of a training sample, the diagnostic model automatically generates a diagnosis result.

B. Feature Extraction Layer

The FE layer is a core module of the proposed method that extracts deep representative features from each fault sample. In the proposed method, the FE layer is composed of several 1-D convolutional layers and 1-D pooling layers.

1) 1-D Convolutional Layer: The 1-D convolution is a special convolution operation. Each 1-D convolutional layer can extract multiple representative features from the 2-D sample [31]. The 1-D convolution operation can be described in the following [22]:

$$C_{l' \times w'}^{(k)} = f \left(W_{l \times w}^{(k)} \otimes X_{t \times w}^{(k)} + B^{(k)} \right) \quad (3)$$

where $\otimes$ represents the convolutional operator; k is k th group 1-D convolution kernel; $W_{l \times w}^{(k)}$ represents the weight of the k th group 1-D convolution kernel; l is the length of 1-D convolution kernel; w represents the width of 1-D convolution kernel and convoluted image; $X_{t \times w}^{(k)}$ is the input image; t represents the length of convoluted image; B is the bias of the k th group 1-D convolution kernel; $C_{l' \times w'}^{(k)}$ represents the output image after 1-D convolution operation; l' is the length of output image of 1-D convolutional layer; w' represents the width of output image; and $f(\cdot)$ is the nonlinear activation function [24].

2) 1-D Pooling Layer: Each 1-D pooling layer can further compress and downsample the dimension and parameter quantity of the previous output feature map [26]. The 1-D pooling layer does not change the width of the image. In CNN, the max pooling is widely used in the pooling layer [29]. The mathematical expression of the 1-D max-pooling can be described in the following:

$$P_{l'' \times w'}^{(k)} = \max_{(j-1)S+1 < t_x \leq jS} \left\{ C_{l' \times w'}^{(k)}(t_x, w') \right\} \quad (4)$$

where x represents the length of 1-D pooling kernel; t_x, w' represents the feature value corresponding to 1-D pooling kernel; S is the stride of 1-D pooling kernel; $P_{l'' \times w'}^{(k)}$ is the output image after 1-D pooling operation; and l'' represents the length of the output image of the 1-D pooling layer.

C. 1-D Global Average Pooling Layer

The 1D-GAP layer is a special function module in the proposed method for reducing the model parameter quantity and diagnosis duration time. The proposed method designs a 1D-GAP layer to replace the FC layer with 2–3 layers of the traditional 1D-CNN model. In the proposed improvement strategy, the 1D-GAP layer has the same role and function as the FC layer. It is used as a transitional bridge that connects the output of the FEL to the input of the logic discrimination layer. The detail of the 1D-GAP layer is shown in Fig. 4. The main principle of the GAP is similar to the pooling operation [5]. The mathematical expression of the 1D-GAP operation can be rewritten as

$$\text{GAP}_{1D}^l = \frac{1}{h} \sum_{i=1}^h X_i^l \quad (5)$$

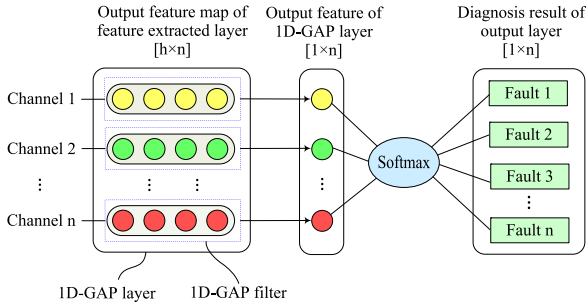


Fig. 4. Schematic diagram of the 1D-GAP operation.

where GAP_{1D}^l represents the calculated result of the l th output channel of the FEL by the 1-D global average pooling filter, l is the index of the output channel, h represents the total number of the elements in the 1D-GAP filter, and X_i^l represents the value of the i th element in the l th channel.

For instance, in Fig. 4, n fault types are shown. It is assumed that the size of the output feature map of the FEL is $[h \times n]$. The main operation process of the 1D-GAP can be summarized as follows. First, the output dimension of the 1D-GAP layer is consistent with the output dimension of the FC layer. In conventional 1D-CNN algorithm, the output dimension of the FC layer is n (n fault type). Therefore, n 1D-GAP filters are designed. Second, the input dimension of the 1D-GAP layer is also consistent with the output size of the FEL. It should be noted that the input dimension of the 1D-GAP layer is equal to the output dimension of the 1D-GAP layer. Because the 1D-GAP operation only compresses the size of the feature map but does not change the number of channels. Therefore, the output dimension of the FE layer should be n . Third, the size of each 1D-GAP filter is set to $[h \times 1]$. Each 1D-GAP filter calculates a global average value from the corresponding channels. After the 1D-GAP layer operation, the output size of the FEL is transformed into $[1 \times n]$. Finally, the output of the 1D-GAP layer ($[1 \times n]$) is fed into the Softmax classifier. It is worth noting that the 1D-GAP layer does not have any training parameters. Therefore, the 1D-GAP layer can greatly reduce the number of training parameters of the CNN model. Thus, the model training time and diagnosis processing time can be reduced substantially.

D. Logic Discrimination Output Layer

In the proposed method, an LDO layer is designed after the 1D-GAP layer. In the LDO layer, there are two classifiers: the Softmax transition classifier and the SVM final classifier. The Softmax transition classifier is used for model training, while the SVM final classifier is used for model testing. It can be seen from Fig. 1 that there are two logical combinations in different stages. In the model training stage, 1D-GAPCNN-Softmax is employed. The Softmax function is first used for error back-propagation and model parameter updating of the FEL and the 1D-GAP layer. In the model testing stage, 1D-GAPCNN-SVM is used. The

fault diagnosis accuracy can be further enhanced by the SVM classifier.

1) Softmax Transition Classifier: The Softmax is used to convert the output result of the 1D-GAP layer to a new result that satisfies a specific probability distribution [5]. Assuming that the output of the 1D-GAP layer is a 1-D array, which has n quantized value, the Softmax function can be defined as

$$Y'_i = \begin{bmatrix} P(y_i = 1|x_i) \\ P(y_i = 2|x_i) \\ \vdots \\ P(y_i = n|x_i) \end{bmatrix}^T = \frac{1}{\sum_{l=1}^n e^{x_i^T w_l}} \begin{bmatrix} e^{x_i^T w_1} \\ e^{x_i^T w_2} \\ \vdots \\ e^{x_i^T w_n} \end{bmatrix}^T \quad (6)$$

where Y'_i is the output of the i th sample after Softmax normalization, the range of Y'_i is from 0 to 1, and $\sum_{l=1}^n y_i^{(l)} = 1$. By comparing the normalized prediction result with the corresponding sample actual label, the cross-entropy loss error can be calculated according to the following [1]

$$J(w) = -\frac{1}{m} \left[\sum_{i=1}^m \sum_{j=1}^n I\{\bar{y}_i = j\} \log \frac{e^{x_i^T w_j}}{\sum_{l=1}^n e^{x_i^T w_l}} \right] \quad (7)$$

where m represents the sample quantity, i represents the i th training sample, j represents the j th category, and $I\{\cdot\}$ is a logical indication function. If the value in the brackets is true, $I = 1$, otherwise $I = 0$, and $J(w)$ is an error loss value [29]. The training process of CNN iteratively adjusts the parameter w to minimize the cost function $J(w)$.

2) SVM Final Classifier: Unlike the traditional CNN where a Softmax classifier serves as the end layer, our model adopts a more robust classifier, SVM [32]. Theoretically, Softmax is only a normalized function, and it does not change the diagnostic accuracy [5]. Compared with the Softmax classifier, SVM can further extract the key representative features from the output of the 1D-GAP layer. In the proposed method, a nonlinear multiclass SVM is used as the backend classifier after the 1D-GAP layer in the model testing stage. Assume that the training dataset is $Z = \{(x_1, y_1), \dots, (x_i, y_i), \dots, (x_n, y_n)\}$, where $x_i \in \chi = R^n$, $y_i \in \psi = \{-1, +1\}$, and $i = 1, 2, \dots, n$, then x_i is the i th input feature vector and y_i is the category label of x_i . Using kernel skills [32], [33] and soft interval maximization techniques, and introducing a slack variable ξ_i , the learning problem of linear indivisible SVM can be described as a soft interval maximization problem [32]

$$\begin{aligned} \min_{w, b, \xi} \quad & \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i \\ \text{s.t.} \quad & y_i(w \cdot x_i - b) \geq 1 - \xi_i \quad \forall (x_i, y_i) \in Z \\ & \xi_i \geq 0, i = 1, 2, \dots, N \end{aligned} \quad (8)$$

where w and b are optimization parameters, ξ_i is a slack variable, and C is a penalty factor. By solving the dual problem of (8) and constructing a suitable Kernel function $K(x \cdot x_j)$ [32], [33], the separation hyperplane that solves the nonlinearity can be

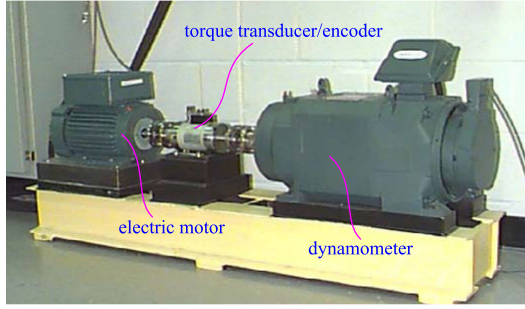


Fig. 5. Rolling bearing fault data acquisition experimental bench.

obtained as follows:

$$\sum_{i=1}^n \alpha_i^* y_i K(x \cdot x_i) + b^* = 0 \quad (9)$$

where α_i^* and b^* are the optimal solutions. This article uses the Gaussian radial basis function (RBF) [32] as the kernel function. The classification decision function using the RBF kernel function can be written as

$$f(x) = \text{sign} \left(\sum_{i=1}^n \alpha_i^* y_i \cdot \exp \left(-\frac{\|x_i - x_j\|^2}{2\sigma^2} \right) + b^* \right) \quad (10)$$

As a mature machine learning technology, SVM has been widely used in pattern recognition tasks. For multiclassification tasks, a multiclass SVM can be constructed by combining multiple binary SVMs [33]. The common combination methods are one-against-one and one-against-all [33], [34]. In this article, we use the one-against-one method.

III. EXPERIMENTAL EVALUATION

To evaluate the effectiveness and performance of the proposed approach for early and fast anomaly diagnosis, two typical multisensor anomaly diagnosis cases, rolling bearing and GPS sensor attack, are investigated in this article.

A. Case 1: Bearing Fault Diagnosis With Multisensors

1) Experimental Setup and Dataset Description: In this case study, a multisensor fault dataset of the rolling bearing from the electromotor is analyzed. The detailed experiment description is explained in [35]. The objective is to diagnose different faults of the bearing with different severity levels. In this experiment, three acceleration sensors are mounted on the drive end, fan end, and base of the motor, respectively, as shown in Fig. 5. In this case study, nine faulty bearing conditions, including the ball defect (BD), outer race defect (OR), and inner race defect (IR), are tested, as shown in Table I. Each fault type has three severity levels. Severity levels are associated with the pitting faults of 0.007 in, 0.014 in, and 0.021 in diameter, respectively [35]. Severity levels indicate gradual evolution of a pitting fault from a microfault to a significant fault. The vibration data of rolling bearing from the three sensors are collected under 1 hp (1772 r/min). The sampling frequency of all data is 12 kHz, while the sampling time is 10 s. Therefore, each sensor can

TABLE I
DESCRIPTION OF THE ROLLING BEARING DATASET UNDER 3-CHANNEL

Fault type	Fault location	Fault size	Fault severity	Raw data length	Total dataset	Training dataset	Testing dataset
Fault 1	BD	0.007	small	[100000, 3]	[900, 500, 3]	630	270
Fault 2	BD	0.014	middle	[100000, 3]	[900, 500, 3]	630	270
Fault 3	BD	0.021	large	[100000, 3]	[900, 500, 3]	630	270
Fault 4	IR	0.007	small	[100000, 3]	[900, 500, 3]	630	270
Fault 5	IR	0.014	middle	[100000, 3]	[900, 500, 3]	630	270
Fault 6	IR	0.021	large	[100000, 3]	[900, 500, 3]	630	270
Fault 7	OR	0.007	small	[100000, 3]	[900, 500, 3]	630	270
Fault 8	OR	0.014	middle	[100000, 3]	[900, 500, 3]	630	270
Fault 9	OR	0.021	large	[100000, 3]	[900, 500, 3]	630	270

TABLE II
HYPERPARAMETERS OF BASELINE REFERENCE 2D-CNN MODEL

Layer type	Kernel	Stride	Activation	Output size	Parameters
Input layer	-	-	-	[batch, 25 × 20, 3]	0
2D-Conv1	[3, 3 × 3, 64]	[1, 1]	Relu	[batch, 25 × 20, 64]	1792
2D-Pool1	[1, 2 × 2, 1]	[2, 2]	-	[batch, 12 × 10, 64]	0
Dropout1	0.3 dropout rate	-	-	[batch, 12 × 10, 64]	0
2D-Conv2	[64, 3 × 3, 32]	[1, 1]	Relu	[batch, 12 × 10, 32]	18464
2D-Pool2	[1, 2 × 2, 1]	[2, 2]	-	[batch, 6 × 5, 32]	0
Flatten	-	-	-	[batch, 960]	0
FC-layer1	128 neuron nodes	-	Tanh	[batch, 128]	123008
Dropout2	0.3 dropout rate	-	-	[batch, 128]	0
FC-layer2	9 neuron nodes	-	Tanh	[batch, 9]	1161
Softmax	Output diagnosis results	-	-	[batch, 9]	0

obtain a long segment of 1-D time-series data with 120 000 data points. In this experiment, in order to facilitate the analysis, the first 100 000 data points are used for fault diagnosis.

In this experiment, the raw 3-channel monitoring data are directly fed into the input layer of the proposed model. As discussed in Section II-A, some data preprocessing operations are performed. First, we can obtain a raw 3-channel 1-D time-series dataset for each fault type, which is a 2-D feature map matrix of [100000 × 3]. Second, according to (2), the long 2-D feature map of [100000 × 3] for each fault type is segmented into 996 short 2-D input feature map of [500 × 3]. Here, the length of the stride is 100. Therefore, the size of each data sample is [500 × 3]. To facilitate analysis, the first 900 samples of each fault type are retained in this experiment. Third, the samples of each fault type are randomly divided into a training dataset (70%) and a testing dataset (30%) [5]. Details of the dataset partition are shown in Table I.

2) Baseline System: In order to verify the diagnosis performance of the proposed method, in this case study, a nearer baseline system of rolling bearing diagnosis model is given, which is based on a 2D-CNN with an FC hidden layer (2D-CNN-FC) [5]. This model consists of two 2D convolution layers, two 2D max-pooling layers, and two FC hidden layers. The number of neurons in each layer is 64, 32, 128, and 9, respectively. All padding mode are "SAME." The detailed hyperparameters of the baseline reference model are given in Table II. This reference baseline model is used to be compared with the proposed method. The input of this 2D-CNN model is the normalized multichannel stacked 2-D feature map. The Softmax function is used as the classifier for supervised learning. The total number of parameters of the 2D-CNN reference model is 144 425.

TABLE III
HYPERPARAMETERS OF THE PROPOSED 1D-GAPCNN-SVM MODEL

Layer type	Kernel	Stride	Activation	Output	Parameters
Input layer	-	-	-	[batch, 500×3]	0
1D-Conv1	[3, 3, 64]	[1,0]	Relu	[batch, 500×64]	640
1D-Pool1	[2, 1, 64]	[2,0]	-	[batch, 250×64]	0
Dropout1	0.3 dropout rate	-	-	[batch, 250×64]	0
1D-Conv2	[3, 64, 32]	[1,0]	Relu	[batch, 250×32]	6176
1D-Pool2	[2, 1, 32]	[2,0]	-	[batch, 125×32]	0
Dropout2	0.3 dropout rate	-	-	[batch, 125×32]	0
1D-Conv3	[3, 32, 9]	[1,0]	Relu	[batch, 125×9]	873
1D-GAP	[125, 1, 9]	[2,0]	-	[batch, 125×9]	0
Softmax	BP training process	-	-	[batch, 9]	0
Input layer	-	-	-	[batch, 500×3]	0
FE layer	Features extraction	-	-	[batch, 125×9]	0
1D-GAP	Dimension transformation	-	-	[batch, 1×9]	0
SVM	Output diagnosis results	-	-	[batch, 9]	90

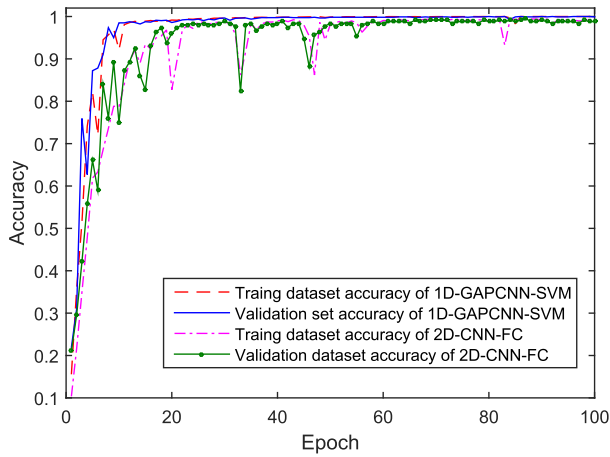


Fig. 6. Accuracy convergence curve of the training process.

3) Diagnosis Model Training and Comparison Experiment: According to the previous research, the anomaly diagnosis model of rolling bearing based on the proposed method is established. The detailed model hyperparameters are listed in Table III. The FE layer is composed of three 1-D convolution layers and two 1-D max-pooling layers. The deep learning training skills, such as data augmentation, dropout technology, and batch normalization, are applied in the model training stage [5]. After each 1-D convolutional layer, batch normalization is used to improve the performance of the proposed model. In this case, the Adam adaptive learning rate optimizer [29] is used. All padding modes are “SAME.”

In this experiment, the number of samples in each minibatch is set to 64. The iteration epoch is 100. Fig. 6 depicts the convergence curve of the model training process. The proposed approach converges within 40 epochs. The fault diagnosis results and the time cost of the traditional 2D-CNN-FC baseline reference model and the proposed 1D-GAPCNN-SVM model are listed in Table IV. In the evaluation system of classifiers, the precision rate (P) and the recall rate (R) are widely used for the performance evaluation of the diagnosis model [32]. The multiclass confusion matrix [1] of the proposed method and the

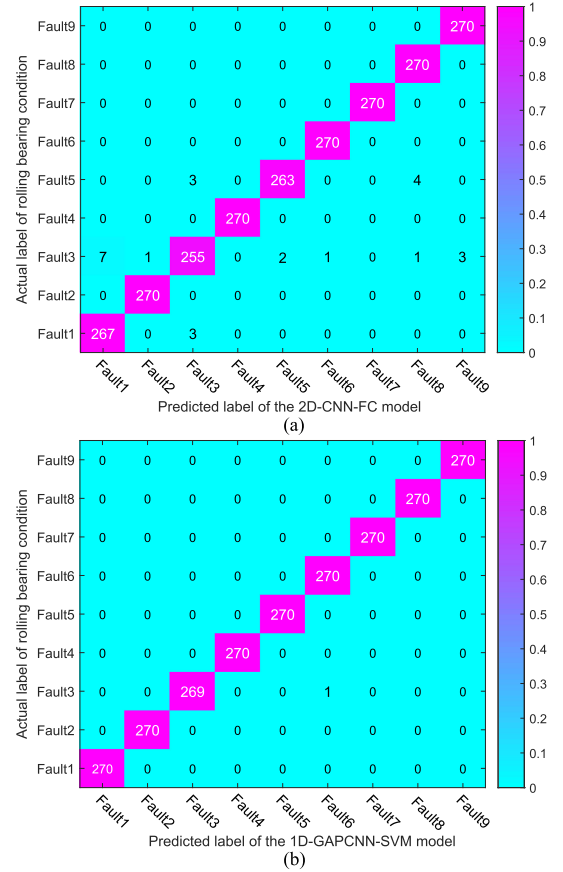


Fig. 7. Multiclass confusion matrix of fault diagnosis results under the different method. (a) Confusion matrix of the traditional 2D-CNN-FC model. (b) Confusion matrix of the proposed 1D-GAPCNN-SVM model.

2D-CNN-FC model corresponding to the results of Table IV are shown in Fig. 7.

To further validate the performance, the proposed method is compared with three conventional machine learning algorithms, including SVM, KNN, and deep BP neural network (DBPN). In conventional data-driven fault diagnosis methods, manual feature extraction is conducted first [24]. Then, the extracted features are fed into the intelligent diagnosis algorithm for classification prediction [5]. Generally, statistical features in time domain and frequency domain are widely used for fault diagnosis [29]. Xia *et al.* [1] extracted 15 features from raw data, including ten time-domain features and five frequency-domain features. In this case study, we use the abovementioned 15 feature extraction operators to calculate the 15 statistical features of each fault sample in Table I. Then, these features are fed into SVM and KNN for fault diagnosis. The outcomes are reported in Table IV.

4) Diagnosis Result Analysis and Discussions: It can be observed from Table IV that the diagnosis performance of the proposed approach is significantly better than the traditional 2D-CNN-FC model and the conventional machine learning models.

First, in terms of accuracy, it can be seen that the deep learning methods are noticeably superior compared to the shallow machine learning algorithms. The precision rates of the

TABLE IV
EVALUATION RESULTS OF DIFFERENT DIAGNOSTIC METHODS

Method name	Conventional 2D-CNN-FC		Transitional 1D-GCNN-Softmax		Proposed 1D-GCNN-SVM		SVM		KNN		DBPN		
Training time (s)	1003.62		846.46		849.74		6.045		—		702.22		
Testing time (s)	1.915		1.312		1.307		1.131		1.933		0.459		
Fault type	P (%)	R (%)	P (%)	R (%)	P (%)	R (%)	P (%)	R (%)	P (%)	R (%)	P (%)	R (%)	Samples
Fault 1	97.45	98.89	100	100	100	100	98.11	95.93	97.47	100	91.10	98.52	270
Fault 2	99.63	100	95.71	99.26	100	100	100	99.63	100	99.26	98.88	97.78	270
Fault 3	97.70	94.44	99.23	95.19	100	99.63	94.74	100	82.24	97.78	91.60	84.81	270
Fault 4	100	100	100	100	100	100	100	100	97.12	100	100	100	270
Fault 5	99.25	97.41	98.53	99.63	100	100	80.36	100	74.92	88.52	98.25	82.96	270
Fault 6	99.63	100	99.63	98.52	99.63	100	83.54	75.19	84.49	58.52	99.26	100	270
Fault 7	100	100	100	100	100	100	100	80.37	100	86.30	99.26	100	270
Fault 8	98.18	100	99.63	100	100	100	96.01	98.15	97.66	92.69	87.38	100	270
Fault 9	98.90	100	100	100	100	100	100	100	92.78	100	98.52	98.52	270
Average/total	98.97	98.97	99.19	99.18	99.96	99.96	94.75	94.36	91.85	91.44	96.03	95.84	5850

conventional 2D-CNN-FC, SVM, KNN, and DBPN on the test dataset are 98.97%, 94.75%, 91.85%, and 96.03%, respectively. However, the accuracy of the proposed 1D-GAPCNN-SVM model is up to 99.96%. It is worth noting that the accuracy of the designed 1D-GCNN-Softmax transition model is 99.19%. This shows that the introduction of SVM as the final classifier successfully improves the accuracy.

In terms of diagnosis speed, the proposed method is much faster than the traditional 2D-CNN-FC model. Training time of the 2D-CNN-FC model is up to 1003.62 s. However, for the case of the proposed model, training time is almost 849.74 s. The training time and testing time are reduced by 15.3% and 31.7%, respectively, in the proposed approach. We notice that the total diagnosis time of the proposed algorithm processing 2430 test samples is 1.307 s (0.0054 s per sample). This detection speed is much more acceptable for fast diagnosis because the mechanical failure is generally a slow process. It can be seen from Table IV, although the shallow machine learning models have shorter training time, the diagnostic accuracy of them is much lower than the proposed model.

In terms of parameter number, the total number of parameters in the baseline reference 2D-CNN-FC model is 144 425. The number of parameters of the FC layer is 124 169, which accounts for 86% of the total parameters. However, the total number of parameters in the proposed 1D-GAPCNN-SVM model is only 7779. The 1D-GAP layer does not have any training parameters. This shows that the 1D-GAP layer can effectively reduce the number of model parameters by more than 86%. Compared with the FC layer, the 1D-GAP layer can reduce the number of training parameters by more than 80%. This indicates that the proposed method enhances the diagnosis speed significantly.

As can be seen from Fig. 7(a), in the 2430 samples, 25 samples are misclassified when the traditional 2D-CNN-FC model is used. The misclassified fault samples mainly contain Fault 1, Fault 3, and Fault 5. In particular, 15 samples of Fault 3 that are misclassified. However, in Fig. 7(b), only one sample is misclassified when the proposed method is used. Especially, the diagnostic accuracy of all incipient microfaults (Faults 1, 4, and 7) is 100%. The experimental results show that the improved 1D-CNN structure and 1D-GAP layer technique can significantly reduce the number of model parameters and the



Fig. 8. Sensors of the autonomous vehicle.

diagnosis time. And the SVM classifier enhances the diagnosis accuracy. Therefore, the proposed method outperforms the previous approaches, in the sense that it is more suitable for fast anomaly diagnosis.

B. Case 2: GPS Anomaly Detection With Multichannel Data

1) Experimental Setup and Dataset Description: In this case study, the proposed method is used for the fast anomaly detection of GPS sensor in automated vehicles (AVs). Autonomous vehicles bring lots of convenience to urban transportation. However, they also face many new challenges such as safety, security, and privacy [36], [37]. To move autonomously, AVs rely on various onboard sensors, such as GPS, cameras, radio detection and ranging, light detection and ranging (LIDAR), and ultrasonic sensors [37], as shown in Fig. 8. However, anomalous sensor readings caused by either malicious cyberattacks or unexpected faults may result in disruptive consequences and possibly lead to fatal crashes [36]. Therefore, it is of great significance to diagnose sensor anomalies in time.

In AV applications, GPS is vulnerable to environmental interferences and malicious attacks. The common anomalous behavior of GPS mainly includes five types [37].

- 1) *Fixed point:* The position measurement is kept constant (by the attacker or due to some technical problems) although the AV moves.

TABLE V
FAULT DESCRIPTION IN GPS CYBERATTACK DATASET

Fault label	Fault type	Fault level	Sample length	Total dataset	Training dataset	Verifying dataset	Testing dataset
0	Normal	No level	[10, 6]	1000	560	140	300
1	Fixed	No level	[10, 6]	1000	560	140	300
2	X-offset	bias: $\pm 1, \pm 2, \pm 4, \pm 6, \pm 8m$	[10, 6]	1000	560	140	300
3	Y-offset	bias: $\pm 5, \pm 10, \pm 15, \pm 20, \pm 25m$	[10, 6]	1000	560	140	300
4	X-gradual drift	ratio: $\pm 0.2, \pm 0.4, \pm 0.6, \pm 0.8, \pm 1.0$	[10, 6]	1000	560	140	300
5	Y-gradual drift	incremental ratio: $\pm 2, \pm 4, \pm 6, \pm 8, \pm 10$	[10, 6]	1000	560	140	300

- 2) *X-direction offset*: The position measurement is affected by a constant bias towards the left or right side of the moving vehicle.
- 3) *Y-direction offset*: The position measurement is affected by a constant bias along the vehicle travel axis towards the front or back.
- 4) *X-direction gradual drift*: The position observation is affected by an incremental drift towards the sides.
- 5) *Y-direction gradual drift*: The position observation is affected by an incremental drift along the travel axis.

Fig. 8 shows a real autonomous driving vehicle of our group. In this vehicle, GPS and LIDAR provide position and orientation measurements in 2-D space, respectively. In this case, 6-channel pose data are combined from GPS and LIDAR for anomaly monitoring. The 3-channel pose data of GPS are anomalous, while the 3-channel pose data of LIDAR are normal. In this case, GPS has six states, including a normal status and five anomalous behaviors abovementioned. The detailed anomaly types and levels are listed in Table V. The sensor sampling frequency of both GPS and LIDAR is 10 Hz. The sampling time of each fault type is 1000 s. To comprehensively consider different anomaly severity levels of GPS, in this experiment, ten anomaly levels are defined for each fault type. Each anomaly level contains 100 s. Each channel contains 10 000 data points. In the input layer, each 6-channel 1-D time-series long data segment of each fault type is fused into a 2-D feature map of $[10000 \times 6]$. This long 2-D feature map of $[10000 \times 6]$ for each fault status is segmented into 1000 short 2-D input feature maps of $[10 \times 6]$.

2) Diagnosis Results and Discussions: A 1D-GAPCNN-SVM structure, similar to the one used in the “rolling-bearing” experiment, is incorporated here too. The filter size of the first convolution layer is adjusted to (3, 6, and 128) due to the change of the data dimension. In this case, parameters are updated using minibatch stochastic gradient descent with a batch size of 64. The iteration epoch is 200. In the training stage, deep learning training skills including data augmentation, dropout technology, and batch normalization are applied. The diagnosis result and

TABLE VI
EVALUATION RESULTS OF TWO DIFFERENT CNN MODELS

Model name	1D-GAPCNN-Softmax		1D-GAPCNN-SVM		
Training time	96.93 (s)		103.19 (s)		
Testing time	0.508 (s)		0.621 (s)		
Health States	Precision	Recall rate	Precision	Recall rate	Samples
Normal state	100%	100%	100%	100%	300
Anomaly 1	100%	100%	100%	100%	300
Anomaly 2	97.61%	95.33%	99.01%	100%	300
Anomaly 3	99.67%	100%	100%	100%	300
Anomaly 4	95.44%	97.67%	100%	99%	300
Anomaly 5	100%	99.67%	100%	100%	300
Average/total	98.79%	98.78%	99.83%	99.83%	1800

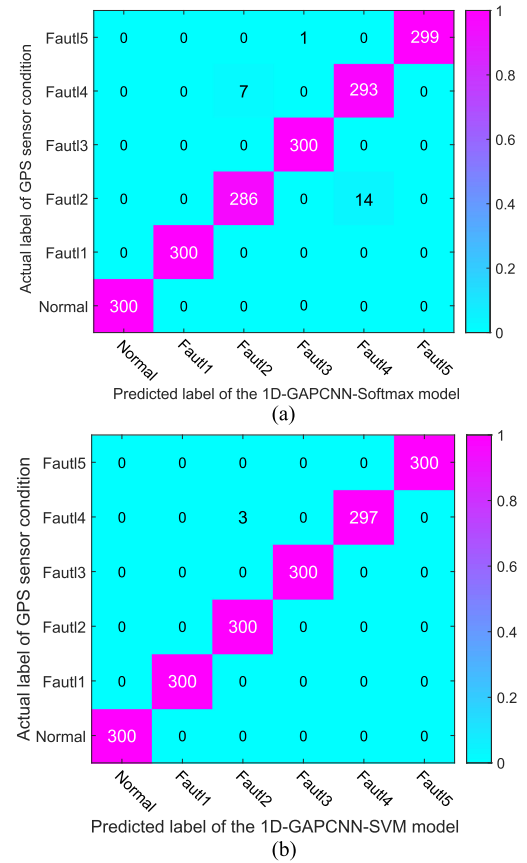


Fig. 9. Multiclass confusion matrix of fault diagnosis results using different methods. (a) Confusion matrix of the 1D-GAPCNN-Softmax model. (b) Confusion matrix of the proposed 1D-GAPCNN-SVM model.

time cost are listed in Table VI. Fig. 9 shows the confusion matrix of the test dataset.

It can be seen from Table VI and Fig. 9 that the proposed method has superior diagnosis performance. In terms of accuracy, the test accuracy of the 1D-GAPCNN-Softmax model is merely 98.79%. However, the final 1D-GAPCNN-SVM model can further enhance accuracy to reach 99.83%. As can be seen from Fig. 9(a), in 1800 samples, 22 samples are misclassified when the 1D-GAPCNN-Softmax model is used. However, in

Fig. 9(b), only three samples are misclassified when the proposed 1D-GAPCNN-SVM diagnosis model is used. In particular, for gradual drift anomaly (Anomaly 4 and Anomaly 5), the identification accuracy of the proposed method is still up to 99%. In terms of time, the total diagnosis time of the proposed algorithm to test 1800 samples is only 0.621 s. Therefore, the experimental results confirm that the proposed 1D-GAPCNN-SVM model demonstrates an outstanding performance for GPS anomaly diagnosis.

IV. CONCLUSION

In this article, a modified 1D-GAPCNN-SVM-based approach with multisensor fusion was proposed for fast anomaly diagnosis. The proposed method mainly addressed three intractable problems of the current 2D-CNN model, including inefficient multisensor data fusion, slower diagnosis speed, and lower diagnosis accuracy. First, 1D-CNN was introduced to address the 1-D time-series data anomaly diagnosis problem. Second, the 1D-CNN was modified by designing a 1D-GAP layer to replace the FC layer with two or three layers, which effectively reduced the number of model parameters. Third, the diagnosis accuracy was further improved using a nonlinear SVM classifier compared with the conventional Softmax classifier. Experiments on rolling bearing and GPS sensor confirmed the superior performance of this method in fault diagnosis. Although the presented case studies were from two different domains, the objective of both cases was multiple classification pattern recognition. Also, the input of both systems was multichannel 1-D time series data. The experimental results showed that the proposed approach had a much faster diagnosis speed and a higher accuracy than the conventional CNN model. The entire diagnosis process in this framework does not require any manual feature extraction and feature selection operations. The end-to-end framework structure of the proposed diagnosis method had superior universality and operability.

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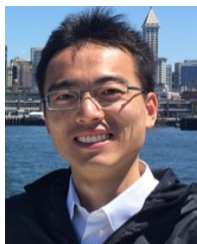
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